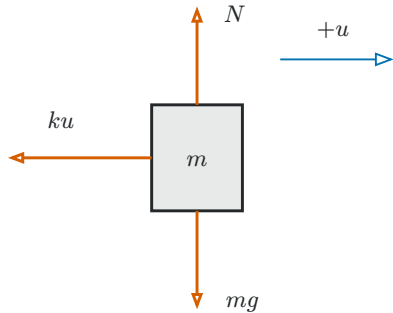
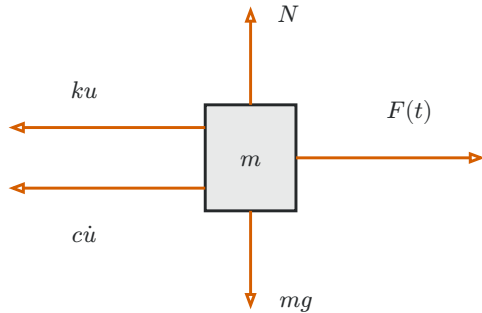


(a) Free, undamped motion



(b) Damping and applied force



Set $c = 0$ for an undamped system; set $F = 0$ for free vibration.

For beam models, N represents the balanced vertical support reaction.